SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

Duration : 45 Minutes

QUESTIONS: 2 **Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

I G. Venkateswarlu *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
   Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

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CO1 - AT3

① Smart Home Automation System using Intelligent Agents.

An Intelligent agent is a system that observes its environment through sensors and take actions to work. In a smart home sensors detect motion, temperature and time. while the system controls lights, air conditioning and security devices.

Agent type

Simple reflex	Responds immediately to sensor inputs	No memory
Model based	Tracks occupancy and environmental conditions	Needs internal state
Goal based	works toward comfort and security goals	Requires Planning

utility based	Balances comfort energy security	More complex

*) A hybrid model-based and utility based would be effective.

*) The model-based component remembers occupancy Patterns and user preferences

*) Simple reflex rules provides notification to emergencies.

Advantages

*) Maintains a comfortable room temperature

*) Ensures all light are off when the house is empty

*) Locks doors when the user leaves home

Conclusion :-

Intelligent agents play an Important role in smart home automation. Simple reflex agents provide quick response. Model-based Maintain environment.

2. Robot Navigation in an unknown Maze using UCS and IDS

Sol) A robot is an unknown maze must explore the environment and find a Path to exit without prior knowledge of the layout.

① Uniform cost search (UCS)

UCS expands the node with the lowest total path cost

① Start from the initial position
② Add reachable to a queue
③ Select the position with the lowest cost.
④ Continue exploring until the exit is reached
⑤ Return the lowest-cost path.

② Iterative Deepening search (IDS)

IDS repeatedly performs depth-limited search, increasing the depth limit after each unsuccessful attempt

① Begin with a depth limit of zero
② Search Paths up to current limit.

③ If the exit is not found, increase the limit

④ Repeat until the exit is reached

The choice of agent and search algorithm affects how efficiently a robot explores an unknown maze.